

# Vendor Performance Analysis and Inventory Optimization

## Exploratory Data Analysis Insights

### Executive Summary

This project focuses on analyzing vendor performance, procurement efficiency, inventory turnover, and profitability using SQL, Python, Statistical Analysis, and Power BI. The analysis was conducted on large-scale supply chain data containing more than 28 million records from sales, purchases, inventory, vendor invoices, and pricing datasets.

The objective was to identify high-performing vendors, evaluate procurement dependency, detect slow-moving inventory, analyze the impact of bulk purchasing, and validate profit margin differences through statistical testing.

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### Project Overview

Effective vendor management is critical for improving profitability, reducing inventory costs, and optimizing procurement strategies. This project aims to provide data-driven insights into vendor performance and inventory management through end-to-end analytics.

#### Business Objectives

- Identify top-performing and low-performing vendors.
- Analyze procurement contribution by vendors.
- Detect brands requiring promotional focus.
- Evaluate inventory turnover efficiency.
- Measure the impact of bulk purchasing on unit costs.
- Validate vendor profitability differences using statistical testing.

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### Dataset Overview

The analysis was performed using the following datasets:

| Dataset         | Number of Records |
|-----------------|-------------------|
| Sales           | 25,650,726        |
| Purchases       | 2,372,474         |
| Begin Inventory | 206,529           |
| End Inventory   | 224,489           |
| Purchase Prices | 12,261            |
| Vendor Invoice  | 5,543             |

Total Records Processed: **28+ Million**

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## Data Engineering Process

The project followed an end-to-end data engineering workflow:

### Step 1: Data Ingestion

Multiple CSV files were imported into a SQL database.

### Step 2: Database Creation

Tables were created for:

- Sales
- Purchases
- Inventory
- Purchase Prices
- Vendor Invoices

### Step 3: Data Processing

Python scripts were used for:

- Data ingestion
- Logging
- Data validation
- Feature engineering

### Step 4: Data Consolidation

SQL joins and aggregations were performed to create a consolidated vendor sales summary table.

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## Feature Engineering

Several business metrics were created for analysis:

### Gross Profit

Measures profitability after accounting for purchase costs.

### Profit Margin

Percentage profit earned on sales.

### Stock Turnover

Measures inventory movement efficiency.

## Sales-to-Purchase Ratio

Compares sales volume against purchased quantity.

## Unit Purchase Price

Average procurement cost per unit.

## Unsold Inventory Value

Estimated capital locked in unsold inventory.

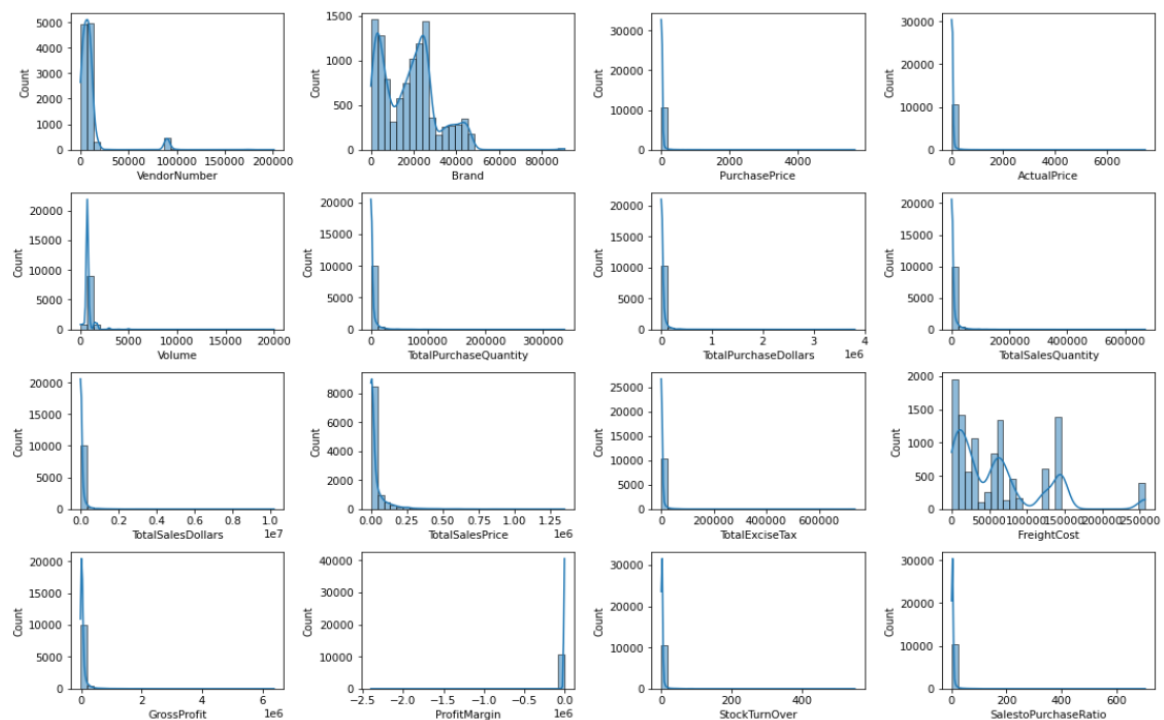
## Unsold Units

Inventory units not sold during the analysis period.

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## Summary Statistics

|                       | count   | mean         | std           | min         | 25%          | 50%          | 75%          | max          |
|-----------------------|---------|--------------|---------------|-------------|--------------|--------------|--------------|--------------|
| VendorNumber          | 10692.0 | 10650.649458 | 18753.519148  | 2.00        | 3951.000000  | 7153.000000  | 9552.000000  | 2.013590e+05 |
| Brand                 | 10692.0 | 18039.228769 | 12662.187074  | 58.00       | 5793.500000  | 18761.500000 | 25514.250000 | 9.063100e+04 |
| PurchasePrice         | 10692.0 | 24.385303    | 109.269375    | 0.36        | 6.840000     | 10.455000    | 19.482500    | 5.681810e+03 |
| ActualPrice           | 10692.0 | 35.643671    | 148.246016    | 0.49        | 10.990000    | 15.990000    | 28.990000    | 7.499990e+03 |
| Volume                | 10692.0 | 847.360550   | 664.309212    | 50.00       | 750.000000   | 750.000000   | 750.000000   | 2.000000e+04 |
| TotalPurchaseQuantity | 10692.0 | 3140.886831  | 11095.086769  | 1.00        | 36.000000    | 262.000000   | 1975.750000  | 3.376600e+05 |
| TotalPurchaseDollars  | 10692.0 | 30106.693372 | 123067.799627 | 0.71        | 453.457500   | 3655.465000  | 20738.245000 | 3.811252e+06 |
| TotalSalesQuantity    | 10692.0 | 6154.964272  | 21905.702782  | 0.00        | 66.000000    | 522.000000   | 3858.500000  | 6.698780e+05 |
| TotalSalesDollars     | 10692.0 | 84478.148838 | 335310.531968 | 0.00        | 1458.440000  | 10596.090000 | 56793.830000 | 1.020384e+07 |
| TotalSalesPrice       | 10692.0 | 37587.567254 | 89905.546771  | 0.00        | 579.420000   | 5715.600000  | 32119.125000 | 1.345639e+06 |
| TotalExciseTax        | 10692.0 | 3548.452518  | 21951.164480  | 0.00        | 9.600000     | 93.140000    | 837.300000   | 7.364856e+05 |
| FreightCost           | 10692.0 | 61433.763214 | 60938.458032  | 0.09        | 14069.870000 | 50293.620000 | 79528.990000 | 2.570321e+05 |
| GrossProfit           | 10692.0 | 54371.455467 | 212941.945045 | -31730.24   | 731.625000   | 6729.745000  | 36298.200000 | 6.392587e+06 |
| ProfitMargin          | 10692.0 | -2597.846393 | 53150.149669  | -2399136.00 | 56.662257    | 65.202729    | 69.978067    | 9.985833e+01 |
| StockTurnOver         | 10692.0 | 3.413587     | 12.040920     | 0.00        | 1.614458     | 1.963058     | 2.078684     | 5.490000e+02 |
| SalestoPurchaseRatio  | 10692.0 | 5.008780     | 16.918135     | 0.00        | 2.307458     | 2.873789     | 3.330898     | 7.058571e+02 |



## Negative & Zero Values

- **Gross Profit:** Minimum value is -31730.24, including losses. Some products or transactions may be selling at a loss due to high costs or selling at discounts lower than the purchase price.
- **Profit Margin:** Has a minimum of -2399136, Which suggests cases where revenue is zero or even lower than costs.
- **Total Sales Quantity & Sales Dollars:** Minimum value are 0, meaning some products were purchased but never sold. These could be slow-moving or obsolete stock

## Outliers Indicated by High Standard Deviations

- **Purchase & Actual Price:** The max values (5681.81 & 7499.99) are significantly higher than the mean (24.39 & 35.64), indicating potential premium products.
- **Freight Cost:** Huge variation, from 0.09 to 257,032.07, suggests logistics inefficiencies or bulk shipments.
- **Stock Turnover:** Range from 0 to 549.0, implying some products sell extremely fast while others remain in stock indefinitely. Value more than 1 indicates that sold quantity for that product is higher than purchased quantity due to sales are being fulfilled from older stock

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## Data Cleaning and Filtering

To improve the reliability of insights, inconsistent records were removed.

The following conditions were applied:

- Gross Profit > 0
- Profit Margin > 0
- Total Sales Quantity > 0

This ensured the analysis focused on active and profitable transactions.

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## Exploratory Data Analysis

Summary statistics and distribution analysis were performed on all key numerical variables.

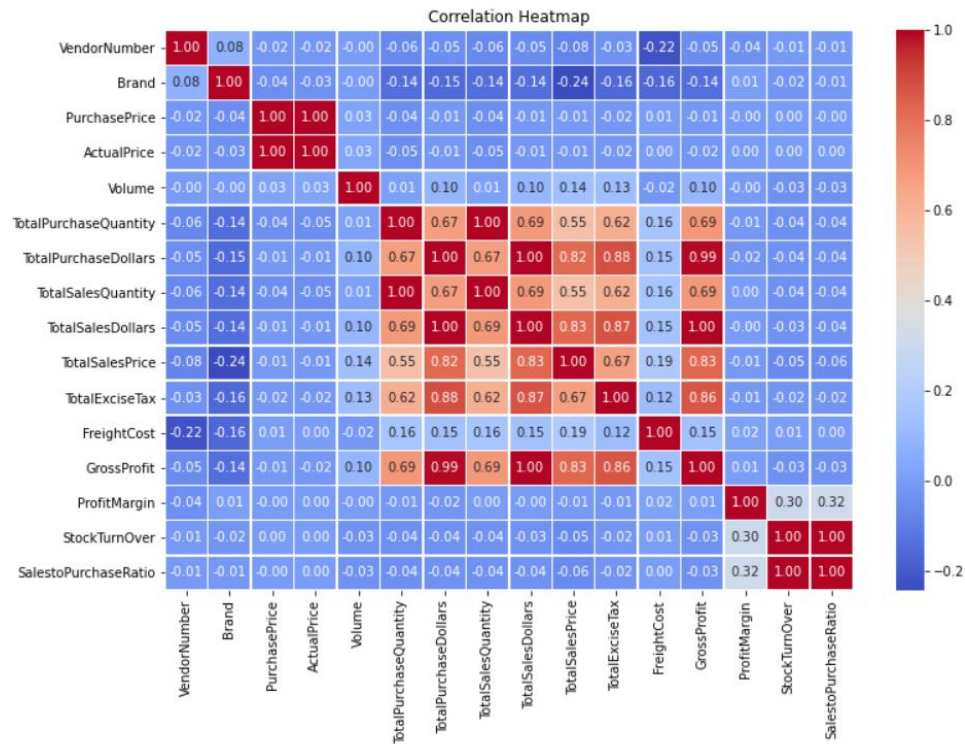
### Key Observations

- Purchase and sales values are highly skewed.
- A small number of vendors contribute significantly to overall revenue.
- Profitability varies considerably across vendors and brands.

- Inventory turnover differs widely between products.

The distribution plots also revealed the presence of outliers, indicating substantial differences in vendor scale and product performance.

## Correlation Analysis



A correlation heatmap was created to understand relationships between business metrics.

### Key Findings

#### Purchase Quantity vs Sales Quantity

Strong positive correlation observed.

#### Insight:

Products purchased in larger quantities generally generate higher sales volumes.

#### Gross Profit vs Total Sales Dollars

Strong positive correlation observed.

#### Insight:

Higher sales revenue contributes directly to increased profitability.

#### Purchase Price vs Profitability

Weak relationship observed.

**Insight:**

Changes in purchase prices alone do not significantly impact profits.

**Stock Turnover vs Profit Margin**

Weak negative correlation observed.

**Insight:**

Faster inventory movement does not necessarily result in higher profitability.

## Research Question 1

Brands with Low Sales but High Profit Margins:

|      | Description                  | TotalSalesDollars | ProfitMargin |
|------|------------------------------|-------------------|--------------|
| 7107 | Santa Rita Organic Svgn Bl   | 19.98             | 83.233233    |
| 2683 | Debauchery Pnt Nr            | 23.16             | 82.987910    |
| 2350 | Concannon Glen Ellen Wh Zin  | 31.90             | 91.724138    |
| 2479 | Crown Royal Apple            | 55.72             | 94.903087    |
| 7147 | Sauza Sprklg Wild Berry Marg | 55.92             | 91.076538    |
| ...  | ...                          | ...               | ...          |
| 3279 | Fifty Shades Of Grey Wh Silk | 901.18            | 94.189840    |
| 6470 | Poggio Amorelli Chianti Clsc | 911.52            | 83.557135    |
| 4283 | JJ Prum Bernkasteler Bads 11 | 917.66            | 84.621755    |
| 5044 | Lincourt Courtney's Chard    | 923.16            | 88.171065    |
| 2120 | Chi Chi's Chocolate Malt RTD | 923.16            | 99.055418    |

186 Brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability



## Which Brands Require Promotional or Pricing Adjustments?

Brands with low sales but high profit margins were identified.

### Finding

Several brands generate strong profit margins despite limited sales volume.

### Business Insight

These brands have growth potential and may benefit from targeted marketing campaigns.

### Recommendation

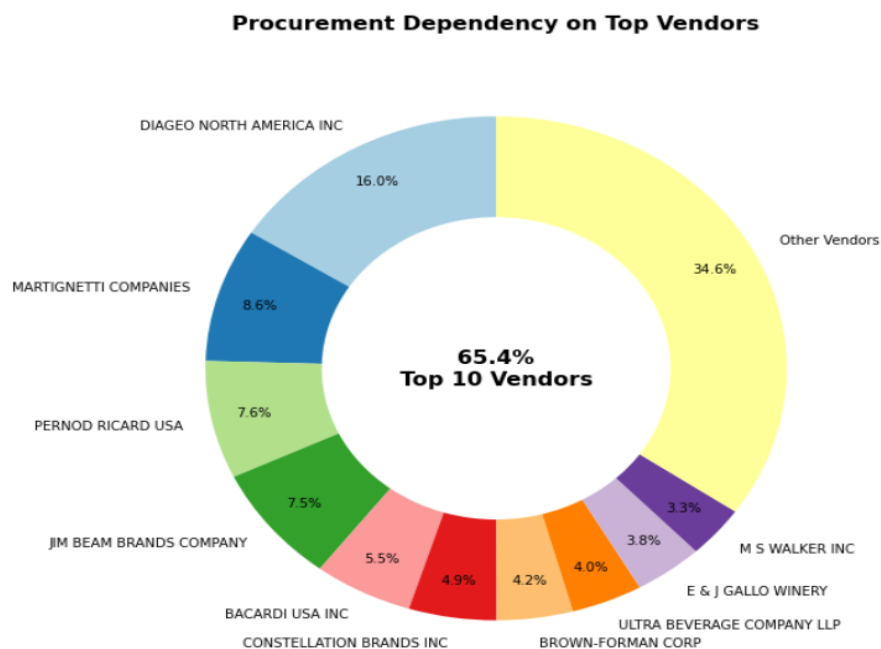
- Improve product visibility.
- Run promotional campaigns.
- Offer bundled discounts.
- Increase distribution reach.

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## Research Question 2

### Procurement Dependency on Top Vendors

The contribution of top vendors toward total purchases was analyzed.



### Finding

The top 10 vendors contribute approximately **65.4%** of total procurement.

### Business Insight

The company relies heavily on a small number of vendors.

#### **Risk**

High vendor concentration may increase supply chain risks.

#### **Recommendation**

Diversify vendor sourcing to reduce dependency and improve supply chain resilience.

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### **Research Question 3**

#### **Impact of Bulk Purchasing on Unit Cost**

The relationship between order size and unit purchase price was analyzed.

##### **Order Size   Average Unit Purchase Price**

Small      39.55

Medium    16.82

Large      11.12

#### **Finding**

Large orders receive significantly lower unit prices.

#### **Key Insight**

Bulk purchases reduce procurement costs by approximately **72%** compared to small orders.

#### **Business Recommendation**

Negotiate bulk purchasing agreements for high-demand products to maximize cost savings.

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### **Research Question 4**

#### **Vendors with Low Inventory Turnover**

Low-performing vendors were identified using Stock Turnover metrics.

#### **Finding**

Several vendors recorded Stock Turnover values below 1.

#### **Unsold Inventory Capital**

Total Unsold Capital:

**\$216.19K**

#### **Business Impact**



Slow-moving inventory:

- Increases storage costs.
- Reduces cash flow efficiency.
- Locks working capital.

### Recommendation

Optimize inventory planning and reduce procurement of slow-moving products.

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## Research Question 5

### Profit Margin Comparison: Top vs Low-Performing Vendors

A confidence interval analysis was performed to compare profitability.

#### Top Vendors

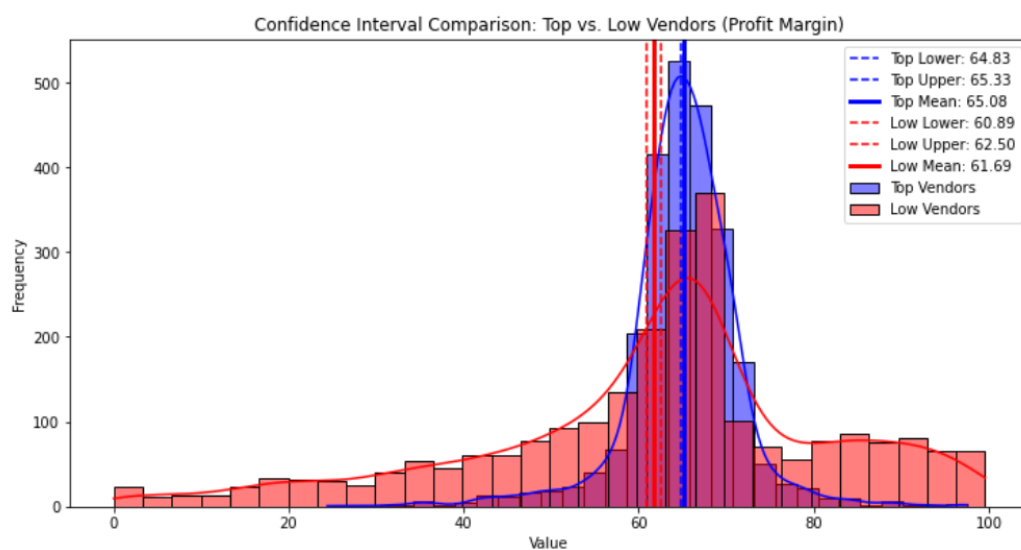
- Mean Profit Margin = 65.08%
- 95% Confidence Interval = (64.83%, 65.33%)

#### Low Vendors

- Mean Profit Margin = 61.69%
- 95% Confidence Interval = (60.89%, 62.50%)

### Finding

Top-performing vendors achieve both higher sales and higher profit margins.



### Business Insight

These vendors demonstrate stronger operational efficiency and product performance.

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## Research Question 6

### Statistical Validation of Profit Margin Differences

A two-sample t-test was conducted.

#### Hypothesis

##### H0 (Null Hypothesis):

There is no significant difference in profit margins between top-performing and low-performing vendors.

##### H1 (Alternative Hypothesis):

The mean Profit margins of top-performing and low-performing vendors are significantly different

#### Test Results

- T-Statistic = 7.8808
- P-Value = 0.0000

#### Conclusion

Since the p-value is less than 0.05, the null hypothesis is rejected.

There is a statistically significant difference in profit margins between top-performing and low-performing vendors.

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## Power BI Dashboard

An interactive Vendor Performance Dashboard was developed in Power BI.

### Dashboard Components

- Total Sales
- Total Purchase
- Gross Profit
- Profit Margin
- Unsold Capital
- Purchase Contribution %
- Top Vendors by Sales
- Top Brands by Sales
- Low Inventory Turnover Vendors

- High Margin Low Sales Brands

The dashboard enables business users to monitor vendor performance and inventory efficiency in real time.

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## Key Business Recommendations

### 1. Focus on Top-Performing Vendors

Strengthen relationships with vendors generating high sales and strong profit margins.

### 2. Promote High-Margin Low-Sales Brands

Increase marketing efforts for brands with strong profitability potential.

### 3. Expand Bulk Purchasing Strategies

Leverage volume discounts to reduce procurement costs.

### 4. Improve Inventory Management

Reduce purchases of slow-moving products and improve turnover.

### 5. Diversify Vendor Base

Reduce procurement dependency on a small number of vendors.

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## Conclusion

This project demonstrates an end-to-end data analytics workflow involving SQL, Python, Statistical Analysis, and Power BI. By analyzing more than 28 million records, valuable insights were generated regarding vendor performance, inventory efficiency, procurement strategy, and profitability. The findings can support better business decisions and contribute to improved operational performance and cost optimization.